RBF-FD Solver: Least Squares Method for Operator Reproduction

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Contents

1 Problem Setting **2**

2 Functional Analysis Aspect **3**

3 RBF-FD Method **4**
3.1 Local Approximation setup 4
3.2 Polynomial Augmentation 4
3.3 Consistency Error Analysis 6
3.4 Growth Function 7

4 Local Least Square Weights RBF-FD **8**
4.1 Weight Constraint Setup 8
4.2 Consistency Error Derivation 8
4.3 Least Squares Finite Spaces and Interpolation Results 9
4.4 Exactness of Approximation on Discrete Spaces 10

5 Appendix **13**

1 Problem Setting

Following the standard RBF-FD methods, let $u \in \mathcal{F} = W^{s,p}$ be a solution for a PDE in form of:

$$-Lu = -\nabla\cdot\big(A\nabla u\big) = f \quad \text{in } \Omega, \tag{1}$$

Throughout this document, we require that L be a self-adjointed, that is, $L = L^*$. Let $1 < p < \infty$, $A \in W^{1,\infty}(\bar{\Omega}; \mathbb{R}^{d\times d}) : \bar{\Omega} \to \mathbb{R}^{d\times d}$ is a symmetric positive definite anisotropic tensor, where

$$\zeta|\xi|^2 \leq \mathbf{A}(\mathbf{x})\xi\cdot\xi \leq \mathbf{Z}|\xi|^{\mathbf{2}},\ \mathbf{0} < \zeta \leq \mathbf{Z} \tag{2}$$

Assume $\Omega \subset \mathbb{R}^d$ and $f \in W^{s-k,p}(\Omega)$ is the forcing term acting in the domain Ω. For 2nd order elliptic operators, we set $k = 2$. The problem is subject to Neumann and Dirichlet Boundary Conditions partitioned within $\partial\Omega$ of the form:

$$u = g_D \quad \text{on } \Gamma_D \subset \partial\Omega, \tag{3}$$
$$n\cdot A\nabla u = g_N \quad \text{on } \Gamma_N \subset \partial\Omega, \tag{4}$$

For simplicity, we assume Ω be a bounded $C^{1,1}(\partial\Omega)$ domain, with $\partial\Omega = \Gamma_D \cup \Gamma_N$ and $\Gamma_D \cap \Gamma_N = \emptyset$. Furthermore, let $g_D \in W^{s-1/p,p}(\partial\Omega)$, $g_N \in W^{s-1-1/p,p}(\partial\Omega)$. By *Agmon-Douglis-Nirenberg* Theorem [2], we have the following estimate for u:

$$\|u\|_{W^{s,p}(\Omega)} \leq C\left(\|Lu\|_{W^{s-k,p}(\Omega)} + \|g_D\|_{W^{s-1/p,p}(\partial\Omega)} + \|g_N\|_{W^{s-1-1/p,p}(\partial\Omega)} + \|u\|_{L^p(\Omega)}\right). \tag{5}$$

In the case that we work on a Hilbert space, we then have for $p = 2$ ($W^{s,2} := H^s$) the following estimate for u:

$$\|u\|_{H^s(\Omega)} \leq C\left(\|Lu\|_{H^{s-k}(\Omega)} + \|g_D\|_{H^{s-\frac{1}{2}}(\partial\Omega)} + \|g_N\|_{H^{s-\frac{3}{2}}(\partial\Omega)} + \|u\|_{L^2(\Omega)}\right).$$

If we want to study cases where $(Lu)(\boldsymbol{x})$ is known pointwise, we can formulate the setup by having $f \in W^{s-k,p}(\Omega)$ and, by Morrey Embedding,

$$W^{s-k,p} \hookrightarrow C^{\mu,\gamma}, \text{ with } \mu := \left\lfloor s - k - \frac{d}{p}\right\rfloor, \gamma := s - k - \mu - \frac{d}{p}$$

In other words, $s > k + \frac{d}{p}$. Suppose $p = 2$ and $k = 2$, then $s > 2 + \frac{d}{2}$. Given an operator $L : H^s(\Omega) \to H^{s-2}(\Omega)$, we define an error functional of the form:

$$\xi_{L,x} : H^s(\Omega) \to \mathbb{R},\ \xi_{L,x} \in (H^s)^*(\Omega)$$

Then by Riesz Representation, there exists a $v_{L,x} \in H^s(\Omega)$ such that $\xi_{L,x}(u) = \langle u, v_{L,x}\rangle_{H^s(\Omega)}$, and $\|\xi_{L,x}|\|_{(H^s(\Omega))^*} = \|v_{L,x}\|_{H^s(\Omega)}$

2 Functional Analysis Aspect

We examine the structural framework that governs the RBF-FD method. Let $\mathcal{N}(\Omega) := W^{s,p}(\Omega)$ be the native space whose vectors are point-wise bounded. We later assume that $\mathcal{N}(\Omega)$ is a Hilbert space (this holds, for instance, when $\mathcal{N}(\Omega)$ is the native space of a symmetric positive definite kernel). Let $L : \mathcal{N}(\Omega) \to W^{s-k,p}(\Omega)$ be a bounded linear operator.
For $n \in \mathbb{N}$, let $\theta_1, \dots, \theta_n \in \mathcal{N}(\Omega)^*$ be bounded linear functionals (for example, point evaluations $\theta_i = \delta_{\boldsymbol{x}_i}$), and let $w_1, \dots, w_n \in \mathbb{R}$ be weight functions. Define $L_w : \mathcal{N}(\Omega) \to W^{s-k,p}(\Omega)$ by

$$L_w u := \sum_{i=1}^{n} w_i \, \theta_i(u). \tag{6}$$

This operator is linear and bounded, since

$$\|L_w u\|_{W^{s-k,p}(\Omega)} \leq \boldsymbol{w}\|_{l^2} \, \|\theta_i\|_{\mathcal{N}(\Omega)^*} \, \|u\|_{\mathcal{N}(\Omega)}.$$

The goal is for L_w to approximate L.

We further assume that there is a linear subspace $\mathcal{V}_h(\Omega) \subseteq \mathcal{N}(\Omega)$ on which L_w reproduces L exactly, i.e., $Lu = L_w u$ for all $u \in \mathcal{V}_w(\Omega)$. We define the error operator $\xi_{w,L} \in \mathcal{L}\big(\mathcal{N}(\Omega), W^{s-k,p}(\Omega)\big)$ by

$$\xi_{w,L}(u) := Lu - L_w u.$$

It is linear and bounded because L and L_w are, and it vanishes on $\mathcal{V}_h(\Omega)$. Let $1 < p < \infty$ and let q satisfy $\frac{1}{p} + \frac{1}{q} = 1$. For $g \in W^{s-k,q}(\Omega)$ define the linear functional l_g on $W^{s-k,p}(\Omega)$ by

$$l_g(v) := \sum_{|\alpha| \leq k} \int_\Omega D^\alpha v \, D^\alpha g \, d\boldsymbol{x}.$$

With $\|v\|_{W^{s-k,p}(\Omega)} = \big(\sum_{|\alpha|\leq k} \|D^\alpha v\|^p_{L^p(\Omega)}\big)^{1/p}$, the Hölder inequality (applied to the integrals and then to the finite sum over α) gives

$$|l_g(v)| \leq \|v\|_{W^{s-k,p}(\Omega)} \|g\|_{W^{s-k,q}(\Omega)}.$$

Set $\xi_{w,L,g} := l_g \circ \xi_{w,L}$. It is linear, and for every $u \in \mathcal{N}(\Omega)$

$$\begin{aligned} |\xi_{w,L,g}(u)| &\leq \|\xi_{w,L}(u)\|_{W^{s-k,p}(\Omega)} \|g\|_{W^{s-k,q}(\Omega)} \\ &\leq \|\xi_{w,L}\|_{\mathcal{L}(\mathcal{N}, W^{s-k,p})} \|g\|_{W^{s-k,q}(\Omega)} \|u\|_{\mathcal{N}(\Omega)}. \end{aligned}$$

Assuming $\xi_{w,L}$ is bounded, then $\xi_{w,L,g}$ is a bounded functional, $\xi_{w,L,g} \in \mathcal{N}(\Omega)^*$. Furthermore, if $\mathcal{N}(\Omega)$ is a Hilbert space, the Riesz Representation Theorem gives a unique $v_{w,L,g} \in \mathcal{N}(\Omega)$ such that $\xi_{w,L,g}(u) = \langle u, v_{w,L,g} \rangle_{\mathcal{N}(\Omega)}$ for all $u \in \mathcal{N}(\Omega)$. Moreover,

$$\begin{aligned} \|\xi_{w,L,g}\|_{\mathcal{N}(\Omega)^*} &= \sup_{\substack{u \in \mathcal{N}(\Omega) \\ \|u\|_{\mathcal{N}(\Omega)} \leq 1}} |\xi_{w,L,g}(u)| \\ &= \sup_{\substack{u \in \mathcal{N}(\Omega) \\ u \neq 0}} \frac{|\xi_{w,L,g}(u)|}{\|u\|_{\mathcal{N}(\Omega)}} \\ &= \sup_{\substack{u \in \mathcal{N}(\Omega) \\ u \neq 0}} \frac{|\langle u, v_{w,L,g} \rangle_{\mathcal{N}(\Omega)}|}{\|u\|_{\mathcal{N}(\Omega)}} = \|v_{w,L,g}\|_{\mathcal{N}(\Omega)}, \end{aligned}$$

3 RBF-FD Method

3.1 Local Approximation setup

To obtain an approximation of the operator L at a point $\boldsymbol{z} \in \Omega$, we approximate u by a function $\hat{u}$ on a neighborhood $\Omega_{\boldsymbol{z}} \subseteq \Omega$ of $\boldsymbol{z}$, so that $Lu(\boldsymbol{z}) \approx L\hat{u}(\boldsymbol{z})$. Let $Y = \{\boldsymbol{y}_1, \ldots, \boldsymbol{y}_m\} \subseteq \Omega_{\boldsymbol{z}}$ be a set of centers, let $\phi_i : \Omega_{\boldsymbol{z}} \to \mathbb{R}$, $i = 1, \ldots, m$, be basis functions, and define the vector space $V_Y = \operatorname{span}\{\phi_1, \ldots, \phi_m\}$. Let $X = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\} \subseteq \Omega_{\boldsymbol{z}}$ be the set of stencil nodes for $\boldsymbol{z}$. For the classical approximation scheme, let $m = n$. We define $\boldsymbol{\Phi} \in \mathbb{R}^{n \times m}$ by $\boldsymbol{\Phi}_{ij} = \phi_j(\boldsymbol{x}_i)$ and assume that $\boldsymbol{\Phi}$ is invertible (for example, $\boldsymbol{y}_i = \boldsymbol{x}_i$ and a strictly positive definite kernel). The role of these basis functions will be to approximate the operator L. We seek $\hat{u} \in V_Y$ of the form

$$\hat{u}(\boldsymbol{x}) = \sum_{i=1}^{n} \lambda_i \phi_i(\boldsymbol{x}) = \boldsymbol{\phi}(\boldsymbol{x})^T \boldsymbol{\lambda}, \qquad \boldsymbol{\phi}(\boldsymbol{x}) := [\phi_1(\boldsymbol{x}), \ldots, \phi_n(\boldsymbol{x})]^T, \tag{7}$$

that interpolates u on the stencil, $\hat{u}(\boldsymbol{x}_i) = u(\boldsymbol{x}_i)$ for $i = 1, \ldots, n$. This leads to the form $\boldsymbol{\Phi}\boldsymbol{\lambda} = \boldsymbol{u}$ for $\boldsymbol{u} = [u(\boldsymbol{x}_1), \ldots, u(\boldsymbol{x}_n)]^T$. We can solve for $\boldsymbol{\lambda}$ to obtain:

$$\hat{u}(\boldsymbol{x}) = \sum_{i=1}^{n} (\boldsymbol{\Phi}^{-1}\boldsymbol{u})_i \, \phi_i(\boldsymbol{x}) = \boldsymbol{\phi}(\boldsymbol{x})^T \boldsymbol{\Phi}^{-1} \boldsymbol{u}.$$

Applying L (which acts on the argument $\boldsymbol{x}$ of $\boldsymbol{\phi}$) and evaluating at $\boldsymbol{z}$, the approximation of Lu at $\boldsymbol{z}$ is

$$Lu(\boldsymbol{z}) \approx L\hat{u}(\boldsymbol{z}) = (L\boldsymbol{\phi})(\boldsymbol{z})^T \boldsymbol{\lambda} = (L\boldsymbol{\phi})(\boldsymbol{z})^T \boldsymbol{\Phi}^{-1} \boldsymbol{u} =: \boldsymbol{w}^{L,X}(\boldsymbol{z})^T \boldsymbol{u}, \tag{8}$$

where $(L\boldsymbol{\phi})(\boldsymbol{z}) := [(L\phi_1)(\boldsymbol{z}), \ldots, (L\phi_n)(\boldsymbol{z})]^T$ and the weights $\boldsymbol{w}^{L,X}(\boldsymbol{z}) \in \mathbb{R}^n$ form a column vector.

Thus, approximating the operator reduces to a local linear solve for the weights. Transposing the definition of $\boldsymbol{w}^{L,X}(\boldsymbol{z})$ gives

$$\boldsymbol{\Phi}^T \boldsymbol{w}^{L,X}(\boldsymbol{z}) = (L\boldsymbol{\phi})(\boldsymbol{z}).$$

The method of undetermined coefficients on the basis $\{\phi_i\}$ gives the same weights: requiring exactness on each basis function, $\sum_{i=1}^{n} w_i \, \phi_j(\boldsymbol{x}_i) = (L\phi_j)(\boldsymbol{z})$ for $j = 1, \ldots, n$, is exactly the system above. Finally, we let the basis of V_Y be defined by radial basis functions, $\phi_i(\cdot) := \varphi(\|\cdot - \boldsymbol{y}_i\|)$, with a radial function $\varphi : [0, \infty) \to \mathbb{R}$ and centers $\{\boldsymbol{y}_i\}_{i=1}^m \subseteq \Omega_{\boldsymbol{z}}$ introduced above. Thus we have defined the Radial Basis Function–generated Finite Difference method (RBF-FD).

Remark. For $\boldsymbol{x} \in \Omega_{\boldsymbol{z}}$ let $w_i(\boldsymbol{x}) := \left(\boldsymbol{w}^{L,X}(\boldsymbol{x})\right)_i$ be the weights computed with the fixed stencil X, and take $\theta_i = \delta_{\boldsymbol{x}_i}$. Then $L_w u = \sum_{i=1}^{n} w_i \, \theta_i(u) = L\hat{u}$ on $\Omega_{\boldsymbol{z}}$, that is, $L_w = L \circ I_X$ where $I_X u := \hat{u}$ is the interpolation operator. If $u \in V_Y$, then $I_X u = u$ and hence $L_w u = Lu$. So the exactness assumption of the previous section holds with $\mathcal{V}_h = V_Y$ (working on $\Omega_{\boldsymbol{z}}$ in place of Ω and assuming $V_Y \subseteq \mathcal{N}$).

3.2 Polynomial Augmentation

For conditionally positive definite kernels, we augment them through the form:

$$\hat{u}(\mathbf{x}) = \sum_{\mathbf{j=1}}^{\mathbf{n}} \lambda_{\mathbf{i}} \phi_{\mathbf{i}}(\mathbf{x}) + \sum_{\mathbf{j=1}}^{\mathbf{l}} \beta_{\mathbf{j}} \mathbf{p}_{\mathbf{l}}(\mathbf{x}) \tag{9}$$

$$\sum_{i=1}^{n} \lambda_i p_j(\boldsymbol{x_i}) = 0, \, \forall i = 1, \ldots, n \tag{10}$$

We can generate a Gram matrix equation as follows:

$$\begin{bmatrix} \hat{\boldsymbol{u}} \\ \boldsymbol{0} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\Phi} & \boldsymbol{P} \\ \boldsymbol{P}^T & \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{\lambda} \\ \boldsymbol{\alpha} \end{bmatrix}$$

Hence, we can approximate the operator as below:

$$(L\hat{u})(\boldsymbol{x}) = \begin{bmatrix} (\boldsymbol{L\phi})(\boldsymbol{x}) \\ (\boldsymbol{Lp})(\boldsymbol{x}) \end{bmatrix}^T \begin{bmatrix} \boldsymbol{\Phi} & \boldsymbol{P} \\ \boldsymbol{P}^T & \boldsymbol{0} \end{bmatrix}^{-1} \begin{bmatrix} \hat{\boldsymbol{u}} \\ \boldsymbol{0} \end{bmatrix} := \begin{bmatrix} \boldsymbol{w}(\boldsymbol{x}) \\ \boldsymbol{\nu}(\boldsymbol{x}) \end{bmatrix}^T \begin{bmatrix} \hat{\boldsymbol{u}} \\ \boldsymbol{0} \end{bmatrix} = \boldsymbol{w}^T(\boldsymbol{x})\hat{\boldsymbol{u}} \tag{11}$$

Thus, we get the local augmented operator reproduction scheme:

$$\begin{bmatrix} (\boldsymbol{L\phi})(\boldsymbol{x}) \\ (\boldsymbol{Lp})(\boldsymbol{x}) \end{bmatrix} = \begin{bmatrix} \boldsymbol{\Phi}^T & \boldsymbol{P} \\ \boldsymbol{P}^T & \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{w}(\boldsymbol{x}) \\ \boldsymbol{\nu}(\boldsymbol{x}) \end{bmatrix} \tag{12}$$

The polynomial constraint setup can be viewed as a Lagrange multiplier to a minimization scheme for the weights. By [6], given the basis for V_Y, $(L\boldsymbol{p})(\boldsymbol{x}) \in \mathit{Range}(\boldsymbol{P}^T) \iff$ Eq. (12) is uniquely solvable, for any set X. Additionally, by [6], Eq. (11) hold $\iff$ Eq. (12) holds. Note that depending on the the basis V_Y, we get the degree of $p \in \Pi^d_{k-1}$. Below are common RBFs and their respective cpd orders: This order provides us both, the degree of the polynomials $k-1$ and the

Table 1: Common radial basis functions and their minimal CPD order.

RBF	**Radial function** $\phi(r)$	**CPD order** k
Gaussian	$\exp(-\varepsilon^2 r^2)$	0
Inverse multiquadric	$(r^2+\varepsilon^2)^{-1/2}$	0
Multiquadric	$(r^2+\varepsilon^2)^{1/2}$	1
Linear	r	1
Cubic	r^3	2
Thin-plate spline	$r^2 \log r$	2
Quintic	r^5	3

number of terms:

$$l = \binom{d+k-1}{k-1}$$

As shown in [8], a polynomial augmentation can be added with a larger degree than required for the RBF, acting as a correction term while inheriting the non-oscillatory behavior of the RBF at the boundary $\partial\Omega$.

3.3 Consistency Error Analysis

We begin by defining the native space of solutions for (1) with (4). Suppose $\mathcal{F}_\mathcal{K} = H^s(\Omega)$ with pointwise bounded evaluations. This defines an RKHS for functions $u \in H^s(\Omega)$, $Lu \in H^{s-k}(\Omega)$. We can also define a *Kernel* $K : \Omega \times \Omega \to \mathbb{R}$ such that:

$$\begin{aligned} f(x) &= \langle f, K(\cdot, x)\rangle_{\mathcal{F}_\mathcal{K}} \\ (Lf)(x) &= \langle f, L_x K(\cdot, \boldsymbol{x})\rangle_{\mathcal{F}_\mathcal{K}} \\ \boldsymbol{w}^T(\boldsymbol{x})\boldsymbol{f} &= \left\langle f, \boldsymbol{w}^T(\boldsymbol{x})\boldsymbol{k}(\boldsymbol{\cdot})\right\rangle_{\mathcal{F}_\mathcal{K}} \\ \xi_{w,L,x}(f) :&= Lf(x) - \boldsymbol{w}^T(\boldsymbol{x})\boldsymbol{f} \end{aligned}$$

According to [5], we have the following proposition:

Proposition 1. *(Davydov, 2025)Let D be a differential operator of order k, and K a cpd kernel of order s on $\Omega \subseteq \mathbb{R}^d$ and $\boldsymbol{X} = \{x_1, \dots, x_n\}$, assume that*

1. *$\partial^{\alpha,\beta} K(\boldsymbol{z}, \boldsymbol{z})$ exists for all $|\alpha|, |\beta| \leq k$*
2. *$L_z K(\boldsymbol{z}, \boldsymbol{x}_i)$ exists for all $i = 1, \dots, N$ and*
3. *The equation 12 is consistent*

Then the kernel-based numerical differentiation formula satisfies:

$$|\xi_{w^*,L,x}| = \left| Lf(\boldsymbol{z}) - \sum_{i=1}^{N} w^*_{z,i} f(\boldsymbol{x_i}) \right| \leq \mathcal{P}^2_{K,L,X}(x)||f||_{\mathcal{F}_\mathcal{K}} \tag{13}$$

We note that L is hence, a functional over $\mathcal{F}_\mathcal{K}$. L_x indicates applying the operator acted on the variable $\boldsymbol{x}$ in the kernel. We define $\boldsymbol{k}(\boldsymbol{z}) = [K(\boldsymbol{z}, \boldsymbol{x}_1), \dots, K(\boldsymbol{z}, \boldsymbol{x}_n)]^T$. Then $\xi_{w,L,x}(f) = \langle f, v_{w,L,x}\rangle$, with $v_{w,L,x} = L_x K(\cdot, \boldsymbol{x}) - \boldsymbol{w}^T(\boldsymbol{x})\boldsymbol{k}(\cdot)$. By assuming point-wise error bounds, $\|\xi_{w,L,x}\|_{\mathcal{F}^*_\mathcal{K}}$, with error estimate $\|\xi_{w,L,x}\|_{\mathcal{F}^*_\mathcal{K}} = ||v_{w,K,L,X,x}||_{\mathcal{F}_\mathcal{K}}$. By [5], the functional is continuous on $\mathcal{F}_\mathcal{K}$ if $L_x K$ exists and both, polynomial consistency and operator exactness are guaranteed. Finally, We define the power function:

$$\mathcal{P}^2_{K,L,X}(x) = \min\left\{ ||v_{w,K,L,X,x}||^2_{\mathcal{F}_\mathcal{K}} : \boldsymbol{w} \in \mathbb{R}^n,\, \xi_{w,L,x} p = 0 \text{ for } p \in \Pi^d_{k-1} \right\} \tag{14}$$

The RKHS norm can be simplified as

$$\begin{aligned} ||v_{w,K,L,X,z}||^2_{\mathcal{F}_\mathcal{K}} &= \langle v_{w,K,L,X,z}, v_{w,K,L,X,z}\rangle_{\mathcal{F}_\mathcal{K}} \\ &= \xi_{w,L,y}\xi_{w,L,x} K(y,x)|_{y=z=x} \\ &= \left\langle L_x K(\cdot, \boldsymbol{z}) - \boldsymbol{w}^T_{\boldsymbol{z}}\boldsymbol{k}(\cdot), L_z K(\cdot, \boldsymbol{z}) - \boldsymbol{w}^T_{\boldsymbol{z}}\boldsymbol{k}(\cdot)\right\rangle_{\mathcal{F}_\mathcal{K}} \\ &= \langle L_z K(\cdot, \boldsymbol{z}), L_z K(\cdot, \boldsymbol{z})\rangle_{\mathcal{F}_\mathcal{K}} - 2\left\langle L_z K(\cdot, \boldsymbol{z}), \boldsymbol{w}^T_{\boldsymbol{z}}\boldsymbol{k}(\cdot)\right\rangle_{\mathcal{F}_\mathcal{K}} + \left\langle \boldsymbol{w}^T_{\boldsymbol{z}}\boldsymbol{k}(\cdot), \boldsymbol{w}^T_{\boldsymbol{z}}\boldsymbol{k}(\cdot)\right\rangle_{\mathcal{F}_\mathcal{K}} \\ &= L_y L_x K(y,x)|_{y=z=x} - 2\sum_{i=1}^{n} w_{i,\boldsymbol{z}} L_x K(\boldsymbol{z}, \boldsymbol{x}_i) + \sum_{i,j=1}^{n} w_{i,\boldsymbol{z}} w_{i,\boldsymbol{z}} K(\boldsymbol{x}_i, \boldsymbol{x}_j) \\ &= L_y L_x K(y,x)|_{y=z=x} - \sum_{i=1}^{n} w_{i,\boldsymbol{z}} L_x K(\boldsymbol{z}, \boldsymbol{x}_i) - \sum_{i=1}^{l} v_i L p_i(\boldsymbol{z}) \end{aligned}$$

(*Need to check this line since [6] cancels the third term.*)

The last two line are carried out by Eq (12).

Now we have an error estimate for functions in $\mathcal{F}_\mathcal{K}$ of the form:

$$\begin{aligned}
|\xi_{w,L,z}(f)| &\leq \|\xi_{w,L,z}\|_{\mathcal{F}_\mathcal{K}^*}||f||_{\mathcal{F}_\mathcal{K}} \\
\inf_{\substack{\boldsymbol{w}\in\mathbb{R}^n \\ \xi(\Pi_{k-1}^d)=0}} |\xi_{w,L,z}(f)| &\leq \inf_{\substack{\boldsymbol{w}\in\mathbb{R}^n \\ \xi(\Pi_{k-1}^d)=0}} \|\xi_{w,L,z}\|_{\mathcal{F}_\mathcal{K}^*}||f||_{\mathcal{F}_\mathcal{K}} \\
|\xi_{w^*,L,z}(f)| &\leq \inf_{\substack{\boldsymbol{w}\in\mathbb{R}^n \\ \xi(\Pi_{k-1}^d)=0}} ||v_{w,K,L,X,z}||_{\mathcal{F}_\mathcal{K}}||f||_{\mathcal{F}_\mathcal{K}} \\
|\xi_{w^*,L,z}(f)| &\leq \mathcal{P}_{K,L,X}(\boldsymbol{z})||f||_{\mathcal{F}_\mathcal{K}}
\end{aligned}$$

3.4 Growth Function

To define the consistency order, we define the following stencil as the stencil set.

$$S_{\boldsymbol{z},X} := \bigcup_{i=1}^{N} [\boldsymbol{z}, \boldsymbol{x}_i]$$

. Then for $U : S_{\boldsymbol{x},X} \times S_{\boldsymbol{x},X} \to \mathbb{R}$, we define a binary Höder continuity semi-norm in the form:

$$|U|_{\boldsymbol{z},X,\gamma} := \sup_{\substack{\boldsymbol{x},\boldsymbol{y}\in S_{\boldsymbol{z},X} \\ \boldsymbol{x}\neq\boldsymbol{z},\boldsymbol{y}\neq\boldsymbol{z}}} \frac{|U(\boldsymbol{x},\boldsymbol{y}) - U(\boldsymbol{x},\boldsymbol{z}) - U(\boldsymbol{z},\boldsymbol{y}) + U(\boldsymbol{z},\boldsymbol{z})|}{\|\boldsymbol{x}-\boldsymbol{z}\|_2^\gamma \|\boldsymbol{y}-\boldsymbol{z}\|_2^\gamma}, 0 < \gamma, \leq 1$$

Then, for $r \in \mathbb{N}$, we get

$$|U|_{\boldsymbol{z},X,r+\gamma} := \frac{1}{(\gamma+1)^2 \dots (\gamma+r)^2} \left(\sum_{|\alpha|=r} \sum_{|\beta|=r} \binom{r}{\alpha} \binom{r}{\beta} |\partial^{\alpha,\beta} U|_{\boldsymbol{z},X,\gamma}^2 \right)^{1/2},$$

with the following coefficient definitions: $\binom{r}{\alpha} = \frac{r!}{\alpha!}, \alpha! = \alpha_1! \dots \alpha_d!$. Growth function is defined as follows:

$$\rho_{k-1,D}(\boldsymbol{z}, X, \mu) := \sup \left\{ Lp(\boldsymbol{z}) : p \in \Pi_{k-1}^d, |p(\boldsymbol{x_j})| \leq \|\boldsymbol{x}_j - \boldsymbol{x}\|_2^\mu, j = 1, \dots, n) \right\} \tag{15}$$

By [9], the dual form of ρ takes the form

$$\rho_{k-1,D}(\boldsymbol{z}, Z, \|\cdot\|) := \inf \left\{ \|\boldsymbol{w}\| : \boldsymbol{w} \in \mathbb{R}^n, \xi_{w,L,x} p = 0, p \in \Pi_{k-1}^d \right\}$$

We finally have the following by [5]:

Theorem 1. *(Davydov, 2025) Let L be a differential operator of order k and K a cpd kernel of order s on Ω. Given $\boldsymbol{z} \in \Omega$ and $\boldsymbol{X} = \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$, with $S_{\boldsymbol{z},\boldsymbol{X}} \subset \Omega$. Assume that 12 is consistent and, for some $r \geq k$ and $\gamma \in (0, 1]$, the kernel K possesses continuous partial derivatives $\partial^{\alpha,\beta} K(\boldsymbol{x}, \boldsymbol{y})$ for all $|\alpha|, |\beta| \leq r$ and all $\boldsymbol{x}, \boldsymbol{y} \in S_{\boldsymbol{z},\boldsymbol{X}}$, and $|K|_{\boldsymbol{z},\boldsymbol{X},r+\gamma} < \infty$. Define $q = \max\{s, r+1\}$. Then for $f \in \mathcal{F}_\mathcal{K}$ the kernel-based numerical differentiation satisfies the error bound:*

$$|\xi_{w^*,L,x}| = \left| Lf(\boldsymbol{z}) - \sum_{i=1}^{N} w_{z,i}^* f(\boldsymbol{x_i}) \right| \leq \rho_{q,L}(\boldsymbol{z}, \boldsymbol{X}, r+\gamma) |K|_{\boldsymbol{z},\boldsymbol{X},r+\gamma} ||f||_{\mathcal{F}_\mathcal{K}} \tag{16}$$

4 Local Least Square Weights RBF-FD

4.1 Weight Constraint Setup

Rather than assuming a form for the solution, our goal is to approximate the operator through weights dual to u sampled at nodes. Suppose $m > n$ (for the case of $m < n$, refer to [1]), thus more center nodes than stencil nodes per stencil. Let $\phi_j(\boldsymbol{z}) = K(\boldsymbol{z}, \boldsymbol{y}_j)$. The following notation for the systems of equations will be used, where X is the stencil nodes and Y is the center nodes:

$$\begin{aligned} K_{XY} &= [K(\boldsymbol{x}_i, \boldsymbol{y}_j)]_{i,j} \\ K_{YX} &= [K(\boldsymbol{x}_i, \boldsymbol{y}_j)]_{j,i} \\ K_{XX} &= [K(\boldsymbol{x}_i, \boldsymbol{x}_j)]_{i,j} \\ K_{YY} &= [K(\boldsymbol{y}_i, \boldsymbol{y}_j)]_{i,j} \end{aligned}$$

Additionally, instead of assuming exactness of the form $L\phi_j(\boldsymbol{x}) = \sum_i^n w_i(\boldsymbol{x})\phi_j(\boldsymbol{x}_i)$ for $j = 1, \ldots, m$ as in the classical RBF-FD, we define a residual function:

$$\begin{aligned} r_j(\boldsymbol{x}) :&= L\phi_j(\boldsymbol{x}) - \sum_{i=1}^{n} w_i(\boldsymbol{x})\phi_j(\boldsymbol{x_i}) \\ r(\boldsymbol{x}) &= L\boldsymbol{\phi}(\boldsymbol{x}) - K_{YX}\boldsymbol{w}(\boldsymbol{x}),\ \boldsymbol{w} : \mathbb{R}^d \to \mathbb{R}^n \text{ for } \boldsymbol{x} \in \Omega. \end{aligned}$$

4.2 Consistency Error Derivation

We use the solution of the interpolation scheme to compare with the ordinary least squares, where $K_{XX}\boldsymbol{w}^* = L\boldsymbol{k} := \boldsymbol{b}$ is the local interpolation scheme. Thus, we can say $\boldsymbol{w}^{*T}K_{XX}\boldsymbol{w}^* = \boldsymbol{b}^T K_{XX}^{-1}\boldsymbol{b}$ since K_{XX} is symmetric. Similar to the interpolation scheme, we define the error analysis and have the form:

$$\begin{aligned} ||v_{w,K,L,X,z}||^2_{\mathcal{F}_\mathcal{K}} &= L_y L_z K(y,z)|_{y=z=x} - 2\sum_{i=1}^{n} w_i(\boldsymbol{x}) L_x K(\boldsymbol{x}, \boldsymbol{x}_i) + \sum_{i,j=1}^{n} w_i(\boldsymbol{x}) w_j(\boldsymbol{x}) K(\boldsymbol{x}_i, \boldsymbol{x}_j) \\ &= L_y L_z K(y,z) - 2\boldsymbol{w}^T\boldsymbol{b} + \boldsymbol{w}^T K_{XX}\boldsymbol{w} \\ &= L_y L_z K(y,z) - 2\boldsymbol{w}^T\boldsymbol{b} + \boldsymbol{w}^T K_{XX}\boldsymbol{w} + \boldsymbol{w}^{*T}K_{XX}\boldsymbol{w}^* - \boldsymbol{b}^T K_{XX}^{-1}\boldsymbol{b} \\ &= L_y L_z K(y,z) - \boldsymbol{b}^T\boldsymbol{w}^* - 2\boldsymbol{w}^T\boldsymbol{b} + \boldsymbol{w}^T K_{XX}\boldsymbol{w} + \boldsymbol{w}^{*T}K_{XX}\boldsymbol{w}^* \\ &= L_y L_z K(y,z) - 2\boldsymbol{b}^T\boldsymbol{w}^* + \boldsymbol{w}^{*T}K_{XX}\boldsymbol{w}^* - 2\boldsymbol{w}^T\boldsymbol{b} + \boldsymbol{w}^T K_{XX}\boldsymbol{w} + \boldsymbol{w}^{*T}K_{XX}\boldsymbol{w}^* \\ &= \mathcal{P}^2_{K,L,X}(\boldsymbol{z}) - \boldsymbol{w}^T K_{XX}\boldsymbol{w}^* - \boldsymbol{w}^{*T}K_{XX}\boldsymbol{w} + \boldsymbol{w}^T K_{XX}\boldsymbol{w} + \boldsymbol{w}^{*T}K_{XX}\boldsymbol{w}^* \\ &= \mathcal{P}^2_{K,L,X}(\boldsymbol{z}) + (\boldsymbol{w} - \boldsymbol{w}^*)^T K_{XX} (\boldsymbol{w} - \boldsymbol{w}^*) \end{aligned}$$

We can now bound the second term by considering $\boldsymbol{w} \to \boldsymbol{w_{LS}}$ obtained from a least squares result:

$$\begin{aligned} K_{XY}K_{YX}\boldsymbol{w_{LS}} &= K_{XY}\boldsymbol{b} \\ K_{XY}K_{YX}\boldsymbol{w_{LS}} - K_{XY}K_{YX}\boldsymbol{w}^* &= K_{XY}\boldsymbol{b} - K_{XY}K_{YX}\boldsymbol{w^*} \\ K_{XY}K_{YX}(\boldsymbol{w_{LS}} - \boldsymbol{w}^*) &= K_{XY}(\boldsymbol{b} - K_{YX}\boldsymbol{w}^*) \\ K_{XY}K_{YX}(\boldsymbol{w_{LS}} - \boldsymbol{w}^*) &= K_{YX}^{\dagger}K_{XY}\boldsymbol{r}^* \\ ||\boldsymbol{w_{LS}} - \boldsymbol{w}^*||_2 &\leq ||K_{YX}^{\dagger}||_2 ||\boldsymbol{r}^*||_2 \\ ||\boldsymbol{w_{LS}} - \boldsymbol{w}^*||_2 &\leq \frac{||\boldsymbol{r}^*||_2}{\sigma_{min}(K_{XY})} \end{aligned}$$

Finally, we say that $r_j^* = v_{w^*,K,L,X,z}(\boldsymbol{y}_j)$. Then $||\boldsymbol{r}^*||_2^2 = ||v_{w^*,K,L,X,z}(Y)||_2^2$. Applying kernel evaluation on $v_{w^*,K,L,X,z}$ and applying the power function for interpolations onto v we get:

$$\begin{aligned}
||\boldsymbol{r}^*||_2^2 &\leq \sum_{i=1}^{m} v_{w^*,K,L,X,z}^2(y_i) \\
&\leq \sum_{i=1}^{m} ||v_{w^*,K,L,X,z}||_{\mathcal{F}_\mathcal{K}} \mathcal{P}_{K,I_X,X}^2(\boldsymbol{y_i}) \\
&\leq \mathcal{P}_{K,L,X}^2(\boldsymbol{z}) \sum_{i=1}^{m} \mathcal{P}_{K,I_X,X}^2(\boldsymbol{y_i})
\end{aligned}$$

Therefore, the error estimate (17) for the local Ordinary Least Squares behaves as follows:

$$\begin{aligned}
||v_{w_{LS},K,L,X,z}||_{\mathcal{F}_\mathcal{K}}^2 &\leq \mathcal{P}_{K,L,X}^2(\boldsymbol{z}) + \lambda_{max}(K_{XX})||\boldsymbol{w} - \boldsymbol{w}^*||_2^2 \\
&\leq \mathcal{P}_{K,L,X}^2(\boldsymbol{z}) + \lambda_{max}(K_{XX}) \frac{\mathcal{P}_{K,L,X}^2(\boldsymbol{z}) \sum_{i=1}^{m} \mathcal{P}_{K,I_X,X}^2(\boldsymbol{y_i})}{\sigma_{min}^2(K_{XY})} \\
&\leq \mathcal{P}_{K,L,X}^2(\boldsymbol{z}) \left(1 + \frac{\lambda_{max}(K_{XX}) \sum_{i=1}^{m} \mathcal{P}_{K,I_X,X}^2(\boldsymbol{y_i})}{\sigma_{min}^2(K_{XY})}\right)
\end{aligned} \tag{17}$$

Suppose $\{p_i\}_{i=1}^l$ is a set of polynomials augmenting the radial basis functions. Let $\Pi_k = span\,\{p_i\}_{i=1}^l$ with polynomial degree up to k. We want to reproduce the operator exactly on the polynomial basis:

$$\begin{aligned}
Lp_j(\boldsymbol{x}) : &= \sum_{i=1}^{n} w_i(\boldsymbol{x}) p_j(\boldsymbol{x_i}) \text{ for } j = 1, \ldots, l \\
L\boldsymbol{p}(\boldsymbol{x}) &= \boldsymbol{P^T w}
\end{aligned}$$

where $\boldsymbol{P}_{ij} := p_j(\boldsymbol{x}_i) \in \mathbb{R}^{n \times l}$. Thus, we have a constrained least squares problem in the form:

$$\begin{cases}
\min J(\boldsymbol{w}(\boldsymbol{x})) := \min \dfrac{1}{2} \|\boldsymbol{r}(\boldsymbol{x})\|_2^2 \quad \text{for } \boldsymbol{x} \in \Omega \\
\text{subject to } K_{YX}\boldsymbol{w} = L\boldsymbol{p}
\end{cases} \tag{18}$$

We can solve this setup by defining the Lagrangian $\mathcal{L}$ for the problem as follows:

$$\begin{aligned}
\mathcal{L}(\boldsymbol{w}, \boldsymbol{\nu}) &:= J(\boldsymbol{w}) + \left\langle \boldsymbol{\nu},\, \boldsymbol{P}^T\boldsymbol{w} - L\boldsymbol{p}(\boldsymbol{x}) \right\rangle \\
&= \frac{1}{2} \left\langle K_{YX}\boldsymbol{w},\, K_{YX}\boldsymbol{w} \right\rangle - \left\langle K_{YX}\boldsymbol{w},\, L\boldsymbol{k} \right\rangle + \frac{1}{2} \left\langle L\boldsymbol{k},\, L\boldsymbol{\phi} \right\rangle \\
&\quad + \left\langle \boldsymbol{\nu},\, \boldsymbol{P}^T\boldsymbol{w} - L\boldsymbol{p}(\boldsymbol{x}) \right\rangle .
\end{aligned}$$

4.3 Least Squares Finite Spaces and Interpolation Results

The resulting KKT equations for this system are then:

$$\begin{aligned}
\nabla_w \mathcal{L} = 0 &: \boldsymbol{\Phi} L\boldsymbol{\phi} = \boldsymbol{\Phi}\boldsymbol{\Phi}^T\boldsymbol{w} + \boldsymbol{P}\boldsymbol{\nu} \\
\nabla_\nu \mathcal{L} = 0 &: \boldsymbol{P}^T\boldsymbol{w} = L\boldsymbol{p}
\end{aligned}$$

The augmented matrix then takes the form:

$$\begin{bmatrix} \boldsymbol{\Phi}\boldsymbol{\Phi}^T & \boldsymbol{P} \\ \boldsymbol{P}^T & \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{w} \\ \boldsymbol{\nu} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\Phi} L\boldsymbol{\phi} \\ L\boldsymbol{p} \end{bmatrix} \tag{19}$$

From [6], Equation (19) has a unique solution given $L\boldsymbol{p} \in Range(\mathbf{P^T})$. The primary goal is to approximate $Lu(\boldsymbol{x}) \approx \boldsymbol{w}^T(\boldsymbol{x})\boldsymbol{u}$, with $\boldsymbol{u} = [u(\boldsymbol{x}_1), \ldots, u(\boldsymbol{x}_n)]^T$. Define $L_h f(\boldsymbol{x}) := \boldsymbol{w}^T(\boldsymbol{x})\boldsymbol{f}$ for some function f. Thus, we have the following:

$$\begin{aligned}
L_h f(\boldsymbol{x}) &= \boldsymbol{w}^T(\boldsymbol{x})\boldsymbol{f} \\
&= \begin{bmatrix} \boldsymbol{w}^T & \boldsymbol{\nu}^T \end{bmatrix} \begin{bmatrix} \boldsymbol{f} \\ \mathbf{0} \end{bmatrix} \\
&= \left(\begin{bmatrix} \boldsymbol{\Phi}\boldsymbol{\Phi}^T & \boldsymbol{P} \\ \boldsymbol{P}^T & \mathbf{0} \end{bmatrix}^{-1} \begin{bmatrix} \boldsymbol{\Phi} L\boldsymbol{\phi} \\ L\boldsymbol{p} \end{bmatrix} \right)^T \begin{bmatrix} \boldsymbol{f} \\ \mathbf{0} \end{bmatrix} \\
&= \begin{bmatrix} L\boldsymbol{\phi}^T\boldsymbol{\Phi}^T & L\boldsymbol{p}^T \end{bmatrix} \begin{bmatrix} \boldsymbol{\Phi}\boldsymbol{\Phi}^T & \boldsymbol{P} \\ \boldsymbol{P}^T & \mathbf{0} \end{bmatrix}^{-T} \begin{bmatrix} \boldsymbol{f} \\ \mathbf{0} \end{bmatrix} \\
&= \begin{bmatrix} L\boldsymbol{\phi}^T\boldsymbol{\Phi}^T & L\boldsymbol{p}^T \end{bmatrix} \begin{bmatrix} \boldsymbol{q} \\ \boldsymbol{\alpha} \end{bmatrix} \\
&= L\boldsymbol{\phi}^T(\boldsymbol{x}) \left(\boldsymbol{\Phi}^T\boldsymbol{q} \right) + L\boldsymbol{p}^T(\boldsymbol{x})\boldsymbol{\alpha}
\end{aligned}$$

Next, we can show that the coefficients $\boldsymbol{q}, \boldsymbol{\alpha}$ relate to the coefficients of u as follows:

$$\begin{bmatrix} \boldsymbol{f} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\Phi}\boldsymbol{\Phi}^T & \boldsymbol{P} \\ \boldsymbol{P}^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{q} \\ \boldsymbol{\alpha} \end{bmatrix}$$

Define $V_h = \left\{ \rho(x) : \rho(x) = \sum_{i=1}^m \lambda_i \phi_i(x) \text{ such that } \lambda \in Range\left(\boldsymbol{\Phi}^T\right) \right\} \subseteq V_Y$. Let $s \in V_h + \Pi_k$. Thus, $s(x) = \boldsymbol{\phi}^T(\boldsymbol{x})\boldsymbol{\Phi}^T q + \boldsymbol{p}^T(\boldsymbol{x})\boldsymbol{\alpha}$ where $\boldsymbol{q} \in \mathbb{R}^n$ and $\boldsymbol{\alpha} \in \mathbb{R}^l$. Then:

$$\begin{aligned}
\boldsymbol{s} = s(X) = \boldsymbol{\Phi}\boldsymbol{\Phi}^T\boldsymbol{q} + \boldsymbol{P}\boldsymbol{\alpha} = f(X) = \boldsymbol{f} \\
\boldsymbol{P}^T\boldsymbol{q} = \mathbf{0}
\end{aligned}$$

Hence, we get interpolation of f at the node set X. Furthermore, from the KKT matrix we can show that $\boldsymbol{\Phi r} = \boldsymbol{P\nu}$ which implies that $\boldsymbol{\Phi r} \in range(\boldsymbol{P}) = \left(ker\left(\boldsymbol{P}^T\right)\right)^{\perp}$. Thus, we get the relationship:

$$\boldsymbol{r}^T \left(\boldsymbol{\Phi}^T\boldsymbol{z} \right) = \mathbf{0} \text{ for } \boldsymbol{P}^T\boldsymbol{z} = \mathbf{0} \tag{20}$$

Finally, we check the consistency error for $s \in V_h + \Pi_k$ with $\boldsymbol{P}^T\boldsymbol{q} = \mathbf{0}$.

$$\begin{aligned}
Ls(x) - L_h s(x) &= L\left(\boldsymbol{\phi}^T(\boldsymbol{x})\boldsymbol{\Phi}^T\boldsymbol{q} + \boldsymbol{p}^T(\boldsymbol{x}) \right) \boldsymbol{\alpha}) - \boldsymbol{w}^T(\boldsymbol{x})\boldsymbol{s} \\
&= L\boldsymbol{\phi}^T(\boldsymbol{x})\boldsymbol{\Phi}^T\boldsymbol{q} + L\boldsymbol{p}^T(\boldsymbol{x})\boldsymbol{\alpha} - \boldsymbol{w}^T(\boldsymbol{x})(\boldsymbol{\Phi}\boldsymbol{\Phi}^T\boldsymbol{q} + \boldsymbol{P}\boldsymbol{\alpha}) \\
&= L\boldsymbol{\phi}^T(\boldsymbol{x})\boldsymbol{\Phi}^T\boldsymbol{q} + L\boldsymbol{p}^T(\boldsymbol{x})\boldsymbol{\alpha} - \boldsymbol{w}^T(\boldsymbol{x})\boldsymbol{\Phi}\boldsymbol{\Phi}^T\boldsymbol{q} - \boldsymbol{w}^T(\boldsymbol{x})\boldsymbol{P}\boldsymbol{\alpha} \\
&= \left(L\boldsymbol{\phi}(\boldsymbol{x}) - \boldsymbol{\Phi}^T\boldsymbol{w}(\boldsymbol{x}) \right)^T \boldsymbol{\Phi}^T\boldsymbol{q} + \left(L\boldsymbol{p}^T(\boldsymbol{x}) - \boldsymbol{P}^T\boldsymbol{w}(\boldsymbol{x}) \right)^T \boldsymbol{\alpha} \\
&= \boldsymbol{r}^T\boldsymbol{\Phi}^T\boldsymbol{q} + \left(L\boldsymbol{p}(\boldsymbol{x}) - \boldsymbol{P}^T\boldsymbol{w}(\boldsymbol{x}) \right)^T \boldsymbol{\alpha}
\end{aligned}$$

$$Ls(x) - L_h s(x) = 0 \text{ due to (19) and (20)} \tag{21}$$

4.4 Exactness of Approximation on Discrete Spaces

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Let u be a solution to (1) and $s \in V_h + \Pi_k$. We can say $u = s + (u - s)$ to get the following:

$$\begin{aligned}
\mathcal{E}[u](\boldsymbol{x}) :&= Lu(\boldsymbol{x}) - L_h u(\boldsymbol{x}) \\
&= Lu(\boldsymbol{x}) - \boldsymbol{w}^T(\boldsymbol{x})\boldsymbol{u} \\
&= L((u-s)+s)(\boldsymbol{x}) - \boldsymbol{w}^T(\boldsymbol{x})\boldsymbol{u} - L_h s(\boldsymbol{x}) + L_h s(\boldsymbol{x}) \\
&= L(u-s)(\boldsymbol{x}) + (Ls(\boldsymbol{x}) - L_h s(\boldsymbol{x})) + L_h(s-u)(\boldsymbol{x}) \\
&= L(u-s)(\boldsymbol{x}) - \boldsymbol{w}^T(\boldsymbol{x})(\boldsymbol{u} - \boldsymbol{s}), \text{ by (21)} \\
&= Le(\boldsymbol{x}) - \boldsymbol{w}^T(\boldsymbol{x})\boldsymbol{e},\ e(\boldsymbol{x}) = u(\boldsymbol{x}) - s(\boldsymbol{x}) \\
&= \mathcal{E}[e](\boldsymbol{x})
\end{aligned}$$

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5 Appendix